Zhiqi Chen

Department of Automation, Tsinghua University, Beijing, China

(+86) 18367487119 zq-chen23@mails.tsinghua.edu.cn zhiqichen05.github.io

Education

- **Tsinghua University** **Beijing, China**
  B.Eng. in Automation *Aug 2023 – Present*
  - GPA: 3.99/4.00 (Rank 1/172, until *Jan 2026*),
  - **A+** in Data Structure, Discrete Mathematics, Random Mathematics and Statistics, Linear Algebra, Computer Languages and Programming, Operations Research, Physics for Scientists and Engineers, etc.
  - Research Advisor: Prof. Gao Huang (LeapLab)

Research Interests

- Reinforcement Learning for Reasoning (LLMs and MLLMs) & Long-Horizon Reasoning and Generalization
- Embodied AI

Research Experience

- **LeapLab, Tsinghua University** **Beijing, China**
  Research Intern *Feb 2025 - present*

  Project: Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?
  - Investigated whether reinforcement learning with verifiable rewards (RLVR) genuinely increases reasoning capacity in large language models beyond their base pretrained models.
  - Conducted substantial large-scale RL experiments, including $pass@k$ evaluation across models and benchmarks, and the perplexity analysis.
  - Demonstrated structural limits of RLVR in expanding intrinsic reasoning capacity, challenging prevailing assumptions about post-training scaling.
  - Co-first author to a NeurIPS 2025 Oral paper (Best Paper Runner-Up, get a unique perfect score 6666).

  Project: Further Project in RL for Reasoning LLMs (Coming Soon)

- **Physical AI Research** **Beijing, China**
  Research Intern *Jul 2025 – Sept 2025*

  Project: Learning Push-Grasp Synergy for Occluded Objects in Cluttered Environments
  - Studied reinforcement learning methods for coordinated push–grasp manipulation in cluttered robotic environments.
  - Bridged the sim-to-real gap by deploying algorithms on physical hardware for experimental validation.
  - Contributed to research accepted at ICRA 2026.

Publications

- **Does Reinforcement Learning Really Incentivize Reasoning Capacity in LLMs Beyond the Base Model?**
  Yang Yue*, **Zhiqi Chen***, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, Gao Huang
  NeurIPS 2025 (Oral, Best Paper Runner-Up Award, perfect score 6666)
  ICML 2025 AI4Math Workshop (Oral, Best Paper Award)

- **Learning Push-Grasp Synergy for Occluded Objects in Cluttered Environments**
  Ziang Li, Haorui Wu, **Zhiqi Chen**, Haozhe Zhang, Yuzhe Huang, Changshui Zhang
  ICRA 2026

Honors & Awards

- National Scholarship of China (2024)
- National Scholarship of China (2025)
- Huang-Yicong Scholarship for AGI, Tsinghua Univ. (2025)
- Zheng-Weimin Scholarship (First Prize), Tsinghua Univ. (2025)

Technical Skills

- **Programming:** Python, C, C++, LaTeX
- **Research Areas:** Reinforcement Learning, LLM Post-training, RLVR, Embodied Manipulation

---

Leadership & Activities

- **THU Tong Class (AGI Class)**
  Student *Feb 2024 – Present*

  – Advanced AI training program focusing on AGI-related research and interdisciplinary scientific training.

- **Men's Football Team, Dept. of Automation, THU**
  Captain *2025 – 2026 Season*
  Player *2024 – 2025 Season*
  Player *2023 – 2024 Season*

  – Competing in THU Football League.